

OLBIA COSTA SMERALDA, Italy

WMO: 165310

Lat: 40.899N

Lon: 9.518E

Elev: 11

StdP: 101.19

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	1.1	2.4	-3.8	2.8	7.1	-2.0	3.2	6.6	13.8	13.0	12.3	12.8	1.8	220	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	34.3	22.4	33.0	22.1	31.8	21.9	25.2	30.4	24.3	29.8	23.5	29.1	5.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	18.8	27.8	22.8	17.6	27.2	21.8	16.5	26.8	77.0	30.3	73.3	29.8	70.1	29.2	32.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.2	9.0	-1.6	37.7	1.1	1.7	-2.4	38.9	-3.0	39.9	-3.7	40.9	-4.5	42.1		
(5)			-2.1	26.5	1.3	1.7	-3.1	27.7	-3.8	28.7	-4.5	29.7	-5.4	30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	9.8	9.8	11.8	14.2	18.0	22.5	25.7	25.6	21.9	18.4	13.9	10.9		
	DBStd	6.26	2.59	2.64	2.61	2.31	2.47	2.74	2.15	2.04	2.51	2.52	2.72	2.72		
	HDD10.0	104	35	33	11	1	0	0	0	0	0	0	3	21		
	HDD18.3	1269	266	239	203	126	36	2	0	0	2	30	136	231		
	CDD10.0	2630	28	27	67	126	249	377	486	485	357	259	119	49		
	CDD18.3	752	0	0	0	1	26	129	228	227	110	31	1	0		
	CDH23.3	7287	0	0	2	9	223	1258	2484	2379	777	151	4	0		
	CDH26.7	2880	0	0	0	0	38	448	1132	1054	192	17	0	0		
(14) Wind	WSAvg	3.8	3.9	3.9	4.4	3.9	3.9	3.8	4.1	3.6	3.5	3.1	3.4	3.8		
(15) Precipitation	PrecAvg	535	52	68	60	51	30	23	4	17	27	67	61	100		
	PrecMax	714	135	234	122	170	134	67	29	52	69	199	152	375		
	PrecMin	385	2	3	2	10	0	1	0	0	1	2	6	4		
	PrecStd	98	35	50	36	37	28	22	7	17	23	60	41	83		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	19.1	22.2	24.8	30.1	35.1	36.7	36.9	32.2	28.8	23.8	19.8		
		MCWB	13.7	13.3	14.5	16.0	18.6	21.3	22.5	22.8	21.6	21.0	17.1	14.8		
	2%	DB	16.9	17.2	20.0	22.6	27.2	32.8	34.2	34.6	30.1	26.2	21.8	18.0		
		MCWB	12.9	12.3	13.2	15.4	17.7	21.4	22.1	23.3	21.3	19.8	16.8	14.1		
	5%	DB	15.8	16.0	18.2	21.0	25.8	30.9	33.0	33.0	28.9	25.0	20.1	16.9		
		MCWB	12.3	11.7	12.7	14.7	17.4	20.6	21.8	22.7	21.0	19.3	16.0	13.6		
	10%	DB	14.8	14.9	17.0	19.8	24.1	29.1	31.9	31.6	27.5	23.8	18.9	15.9		
		MCWB	11.4	11.0	12.5	14.3	17.0	20.0	21.9	22.4	20.6	18.7	15.3	12.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	14.2	15.9	17.7	20.4	24.5	26.2	27.1	24.7	22.6	18.7	15.9		
		MCDB	17.2	17.5	20.2	22.0	25.6	30.7	31.5	30.9	28.2	26.1	21.2	18.2		
	2%	WB	13.6	13.1	14.6	16.5	19.5	23.0	24.9	25.7	23.4	21.2	17.6	14.9		
		MCDB	16.1	16.2	18.0	20.6	24.5	29.3	30.6	30.3	27.5	24.4	20.4	17.2		
	5%	WB	12.6	12.3	13.7	15.7	18.6	22.1	24.1	24.7	22.5	20.3	16.7	14.0		
		MCDB	15.2	15.2	16.9	19.5	23.6	28.0	29.8	30.0	26.7	23.6	19.4	16.3		
	10%	WB	11.6	11.4	12.9	15.0	17.8	21.1	23.3	23.8	21.6	19.4	15.6	13.1		
		MCDB	14.2	14.2	16.2	18.4	22.6	27.0	29.1	29.5	25.8	22.8	18.3	15.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.0	9.6	10.2	11.0	11.7	11.8	11.6	10.4	10.0	8.6	8.2		
		MCDBR	9.5	10.7	11.7	12.5	13.3	14.0	13.5	14.0	11.7	11.6	10.4	9.2		
	5% WB	MCWBR	6.5	6.9	7.0	7.0	6.4	6.4	6.0	7.1	5.8	6.5	6.6	6.2		
		MCWBR	9.0	9.9	10.5	11.2	12.0	12.7	12.5	12.8	11.4	11.0	9.8	8.5		
(40) Clear-Sky Solar Irradiance	taub		0.336	0.354	0.384	0.417	0.430	0.412	0.417	0.414	0.410	0.392	0.368	0.340		
		taud	2.473	2.409	2.335	2.255	2.259	2.349	2.346	2.364	2.371	2.398	2.407	2.456		
	Ebn at Noon		817	853	863	856	851	865	856	849	827	798	765	777		
		Edh at Noon	78	95	114	132	135	123	123	117	109	95	83	74		
(44) All-Sky Solar Radiation	RadAvg		1.82	2.68	3.93	5.13	6.22	7.17	7.36	6.38	4.74	3.31	2.05	1.63		
	RadStd		0.19	0.26	0.43	0.37	0.53	0.31	0.30	0.34	0.20	0.29	0.16	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.31	N/A	N/A	N/A	-0.68	-0.94	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page